Split RIR Reverb System

"A hybrid reverb system combining impulse response convolution (early reflections) and algorithmic reverberation (FDN)."

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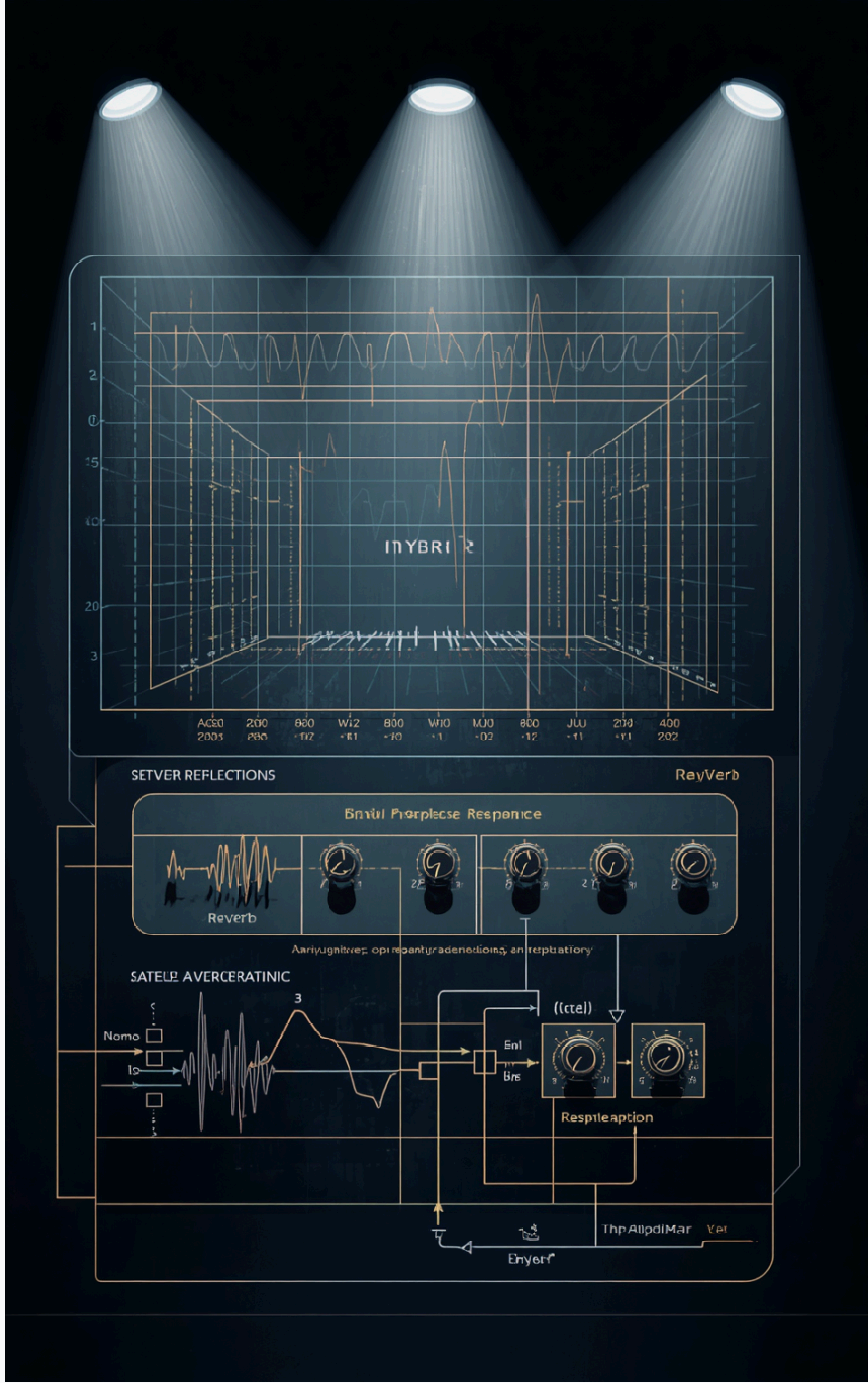
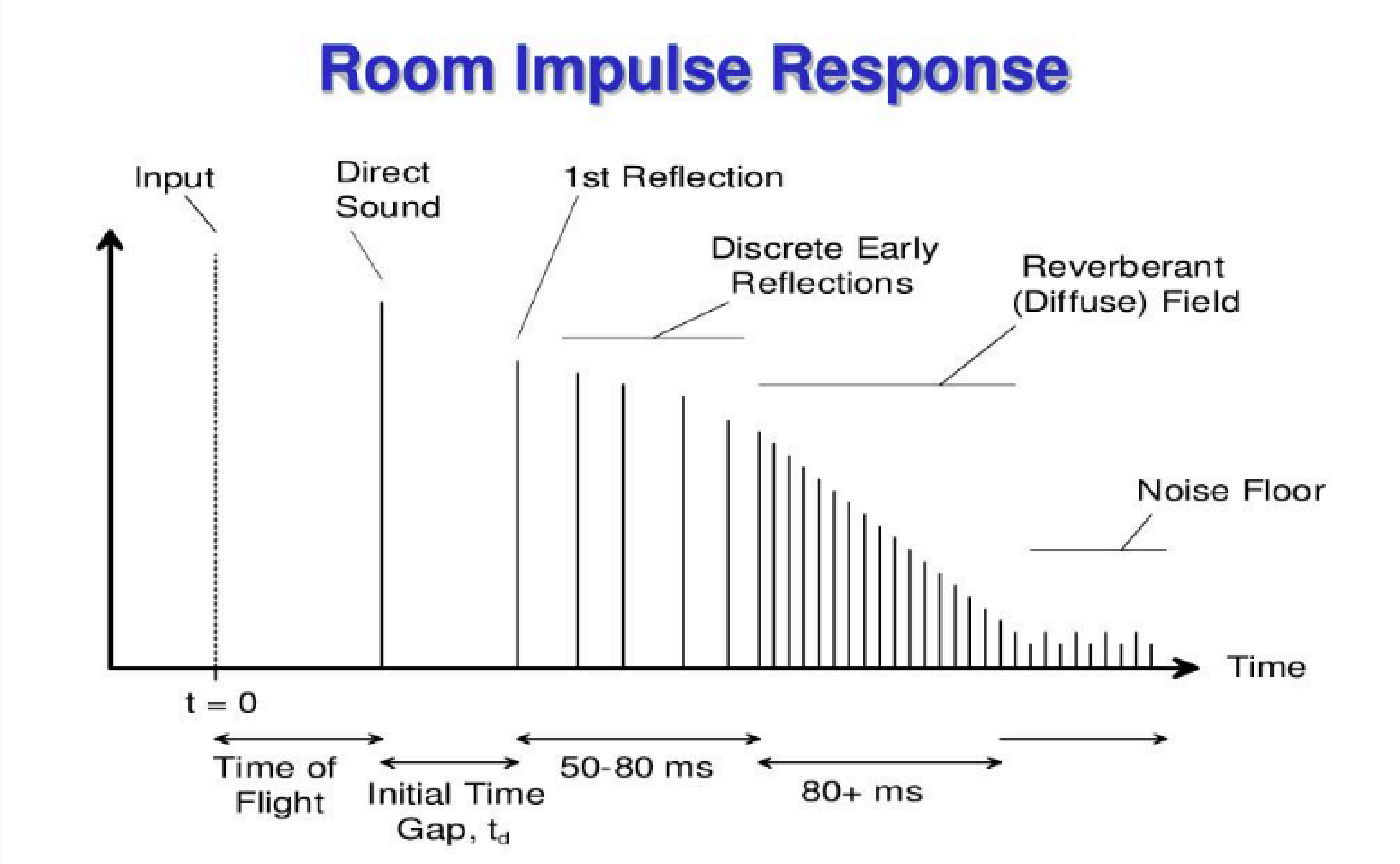


Core Principle

Understanding the split RIR for improved efficiency for real-time system



From https://www.slideserve.com/duman/sound-system-calibration

Computational Cost

Full convolution applies the measured RIR, but **computational costs rise** significantly when IR length increases, affecting real-time applications.

Early Reflections

The proposed method retains perceptually important **early reflections** as convolution reverb, to keep spatial detail and localization cues.

Algorithm Reverb Tail

The late tail part uses an **algorithmic reverb network** to approximate the decay tail cuz the late tail is more diffused and stochastic.

Early Reflection Implementation

Direct Sound Detection

Direct sound detection is based on **sliding short-window RMS** energy. The threshold is:

$$T = \mu_{\text{noise}} + k \cdot \sigma_{\text{noise}}$$

The first frame above T is treated as the direct-sound arrival time t0.

Early Reflection Detection

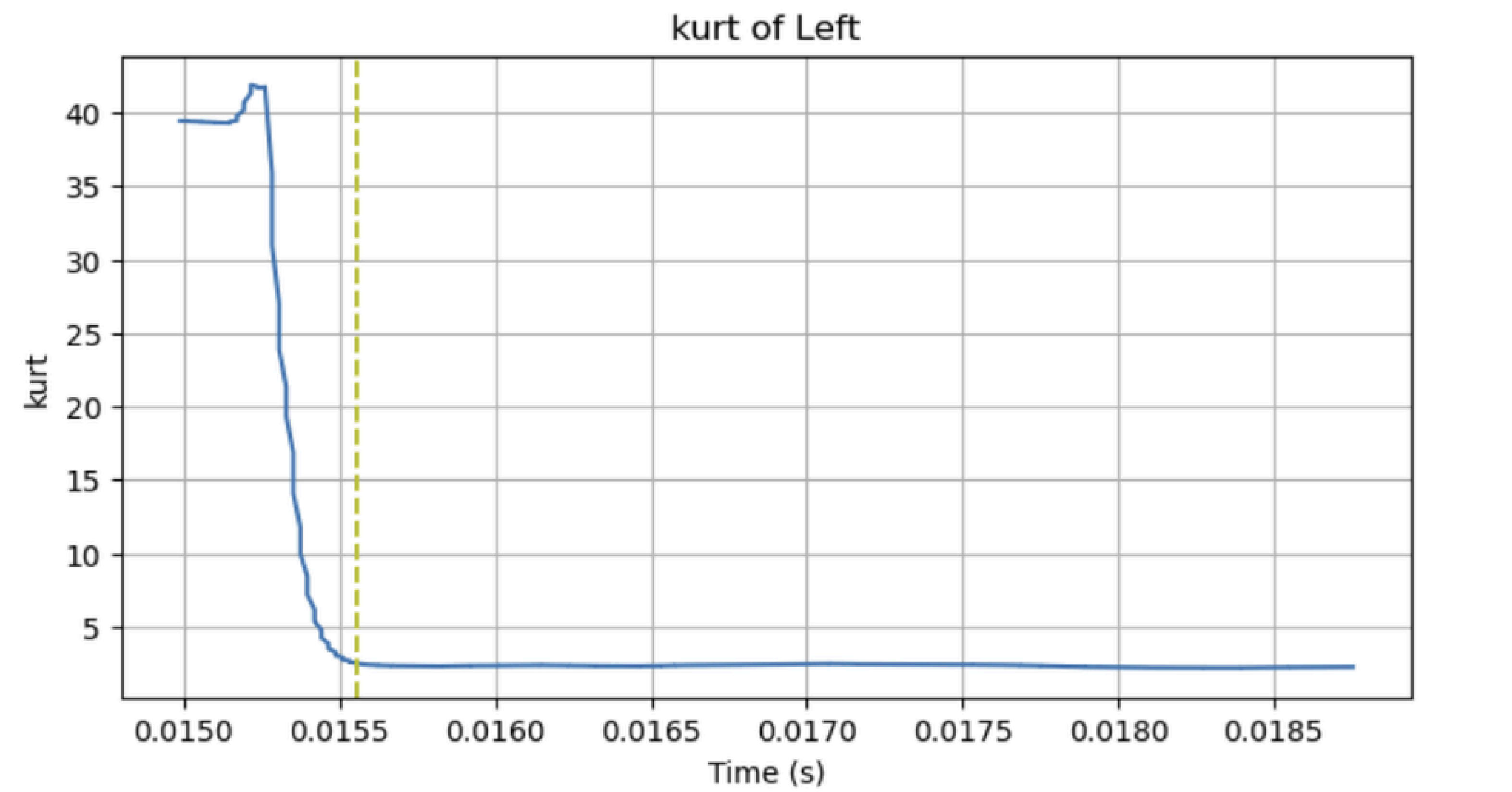
The early/late boundary is estimated using **sliding-window kurtosis** cuz late tail is close to the **Gaussian distribution**:

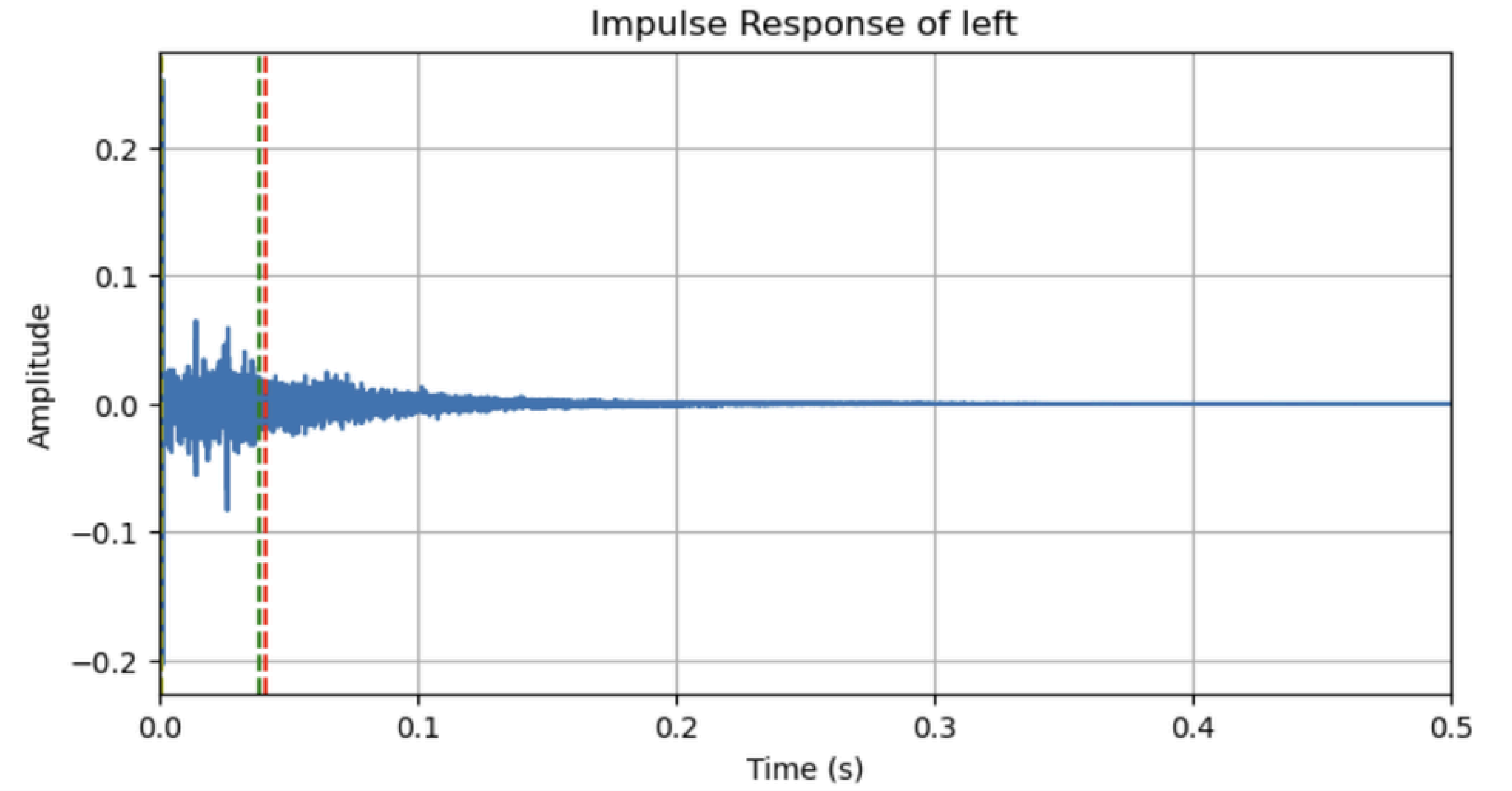
$$\frac{1}{N}\sum_{i=1}^{N}\left(\frac{x_i-\mu}{\sigma}\right)^4 - 3$$

And find the strongest transition of kurt curve asplit point.

RFFT Convolution

The early impulse response is processed using **block-based RFFT convolution**, enabling high-quality stereo output while preserving localization and spatial impression.





Late Tail Implementation

Parameter Extraction

Parameters for the late IR are extracted rather than convolving the entire tail, optimizing processing efficiency while maintaining quality.

RT60 Estimation

RT60 is estimated from the **Energy Decay Curve (EDC)** by performing linear regression on the decay between -5 dB and -35 dB, then extrapolating the fitted line to -60 dB.

Damping Behavior

Frequency-dependent damping is mapped from the **low/high-pass IRs' RT60 difference**:
Delta_RT60 = RT60_low - RT60_high

$$d = d_{\min} + (d_{\max} - d_{\min})\left(1 - e^{-\Delta RT_{60}}\right)$$

Decay Gain

4*4 FDN feedback gain is computed per delay line using estimated rt60:

$$g = 10^{\frac{-3n}{RT_{60} \cdot F_s}}$$

This maps the desired decay time onto each branch length n.

Stereo Width

Stereo width uses **mid-side energy** with imbalance compensation:

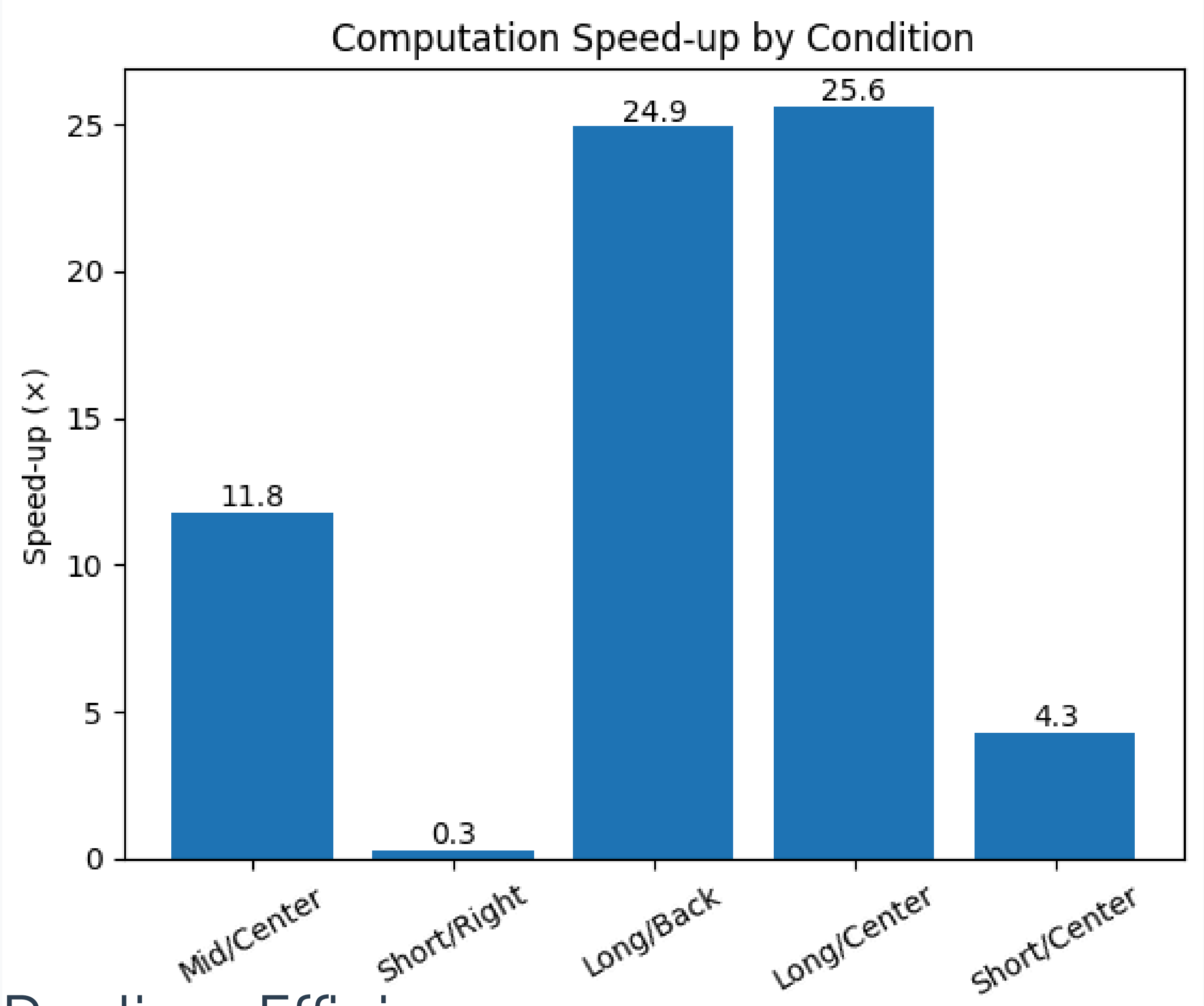
$$\text{width} = \frac{E_S}{E_M + E_S} \cdot \left(1 - \left|\frac{E_R - E_L}{E_L + E_R}\right|\right)$$

And width set up thresholds for different delay sample numbers for allpass filters and delay taps to increase diffusion

Evaluation Results

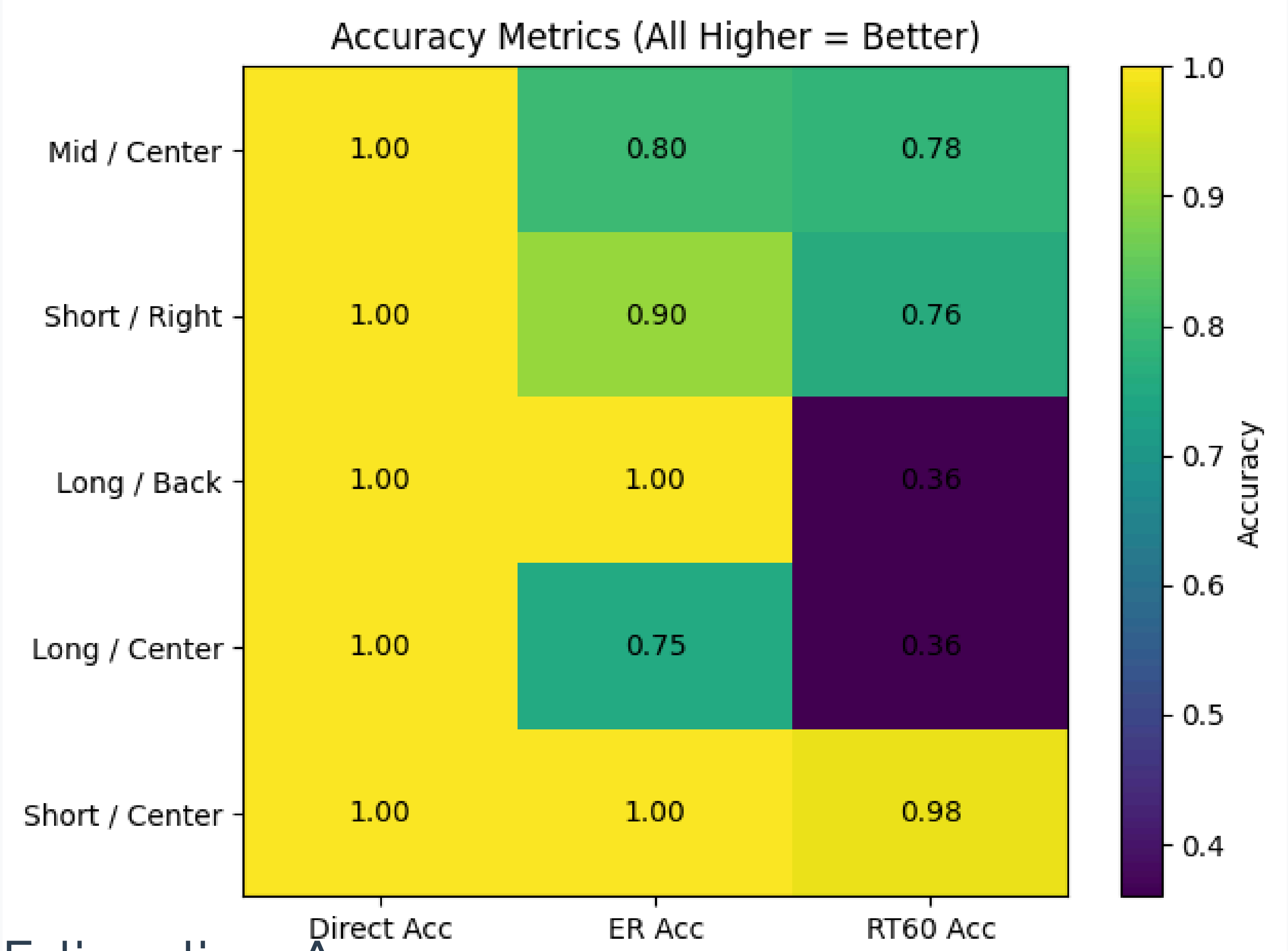
Computation and Accuracy Measurements

Measure the accuracy of the estimated values using a normalized score in the range [0, 1] and improvements of computation speeds.



Runtime Efficiency

The hybrid method achieved approximately 25 times faster processing for long reverb cases, but much lower for short reverb.



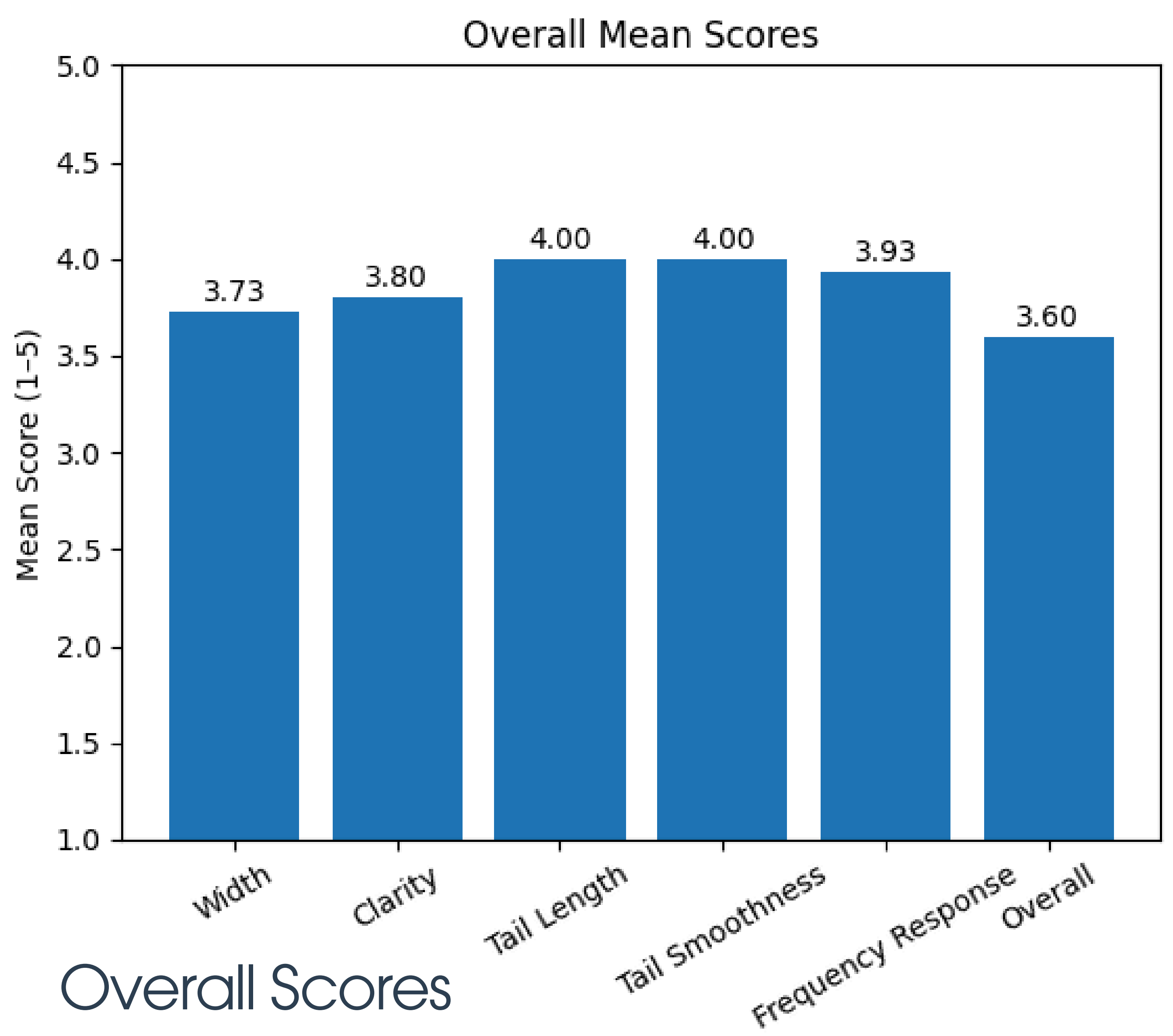
Estimation Accuracy

Early reflections are generally accurate (within the first 50–80 ms), while RT60 accuracy drops significantly in multi-slope EDC conditions, especially for longer IR.

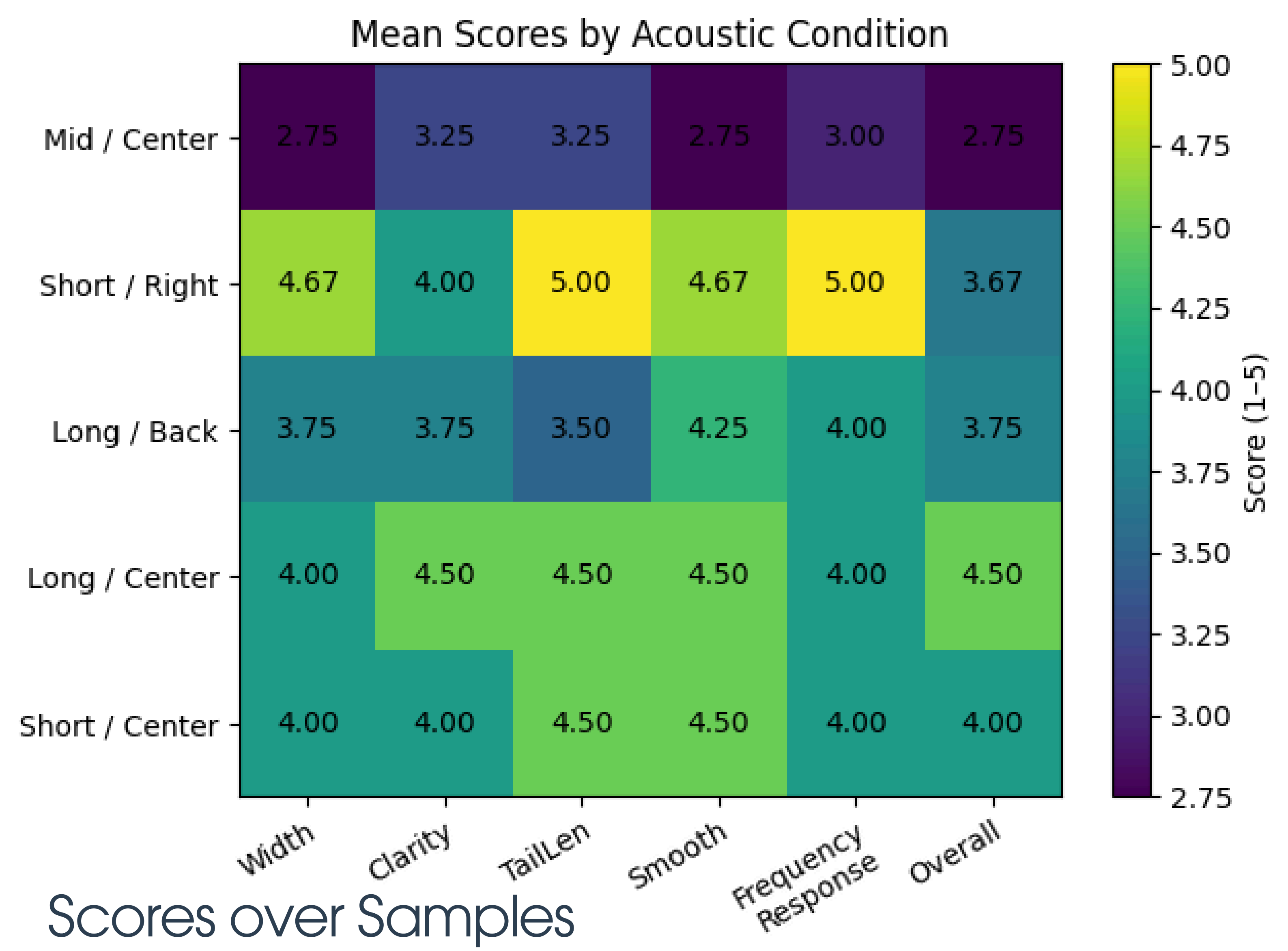
Evaluation Results

Subject Listening

Compare the full convolved audio with the hybrid reverbed audio of their similarities and nuances by 5-point scale evaluation.



Overall Scores



Scores over Samples

The overall scores of each evaluation metric are between 3.5 and 4.0, showing the system has a measure of perceptual acceptability.

The mean scores of long IRs and short IRs are around 4, whereas the mean scores of mid-length IRs are much lower, showing the sample dependency.

Discussion & Improvements

Efficiency vs. Accuracy

The project reveals a **trade-off**; long RIRs benefit most from hybrid models, balancing efficiency and perceptual quality effectively.

RT60 Limitations

Multi-slope Energy Decay Curves complicate RT60 fitting, especially in non-ideal diffusion fields, affecting the accuracy of reverberation tail models.

Late-Tail Modeling

Improvements in late-tail modeling can arise from **multi-band** decay estimation and enhanced **diffusion techniques** for more natural reverb tails.

Future Enhancements

Future work should incorporate **partitioned convolution** techniques for improved real-time performance and overall system responsiveness in reverberation.

Reducing Empirical Params

Future work should involve AI methods like **machine learning**, especially for FDN late reverb tail, to avoid frequent use of preset params

Thank you for your attention!

github: https://github.com/xx2644-xiaoyixu/Split-RIR-Reverb

Reference:

- *Schroeder, M. R. (1962). Natural sounding artificial reverberation. Journal of the Audio Engineering Society, 10.*
- *Schroeder, M. R. (1965). New Method of Measuring Reverberation Time. The Journal of the Acoustical Society of America, 37(6 Supplement), 1187-1188. https://doi.org/10.1121/1.1939454*
- *Dattorro, Jon; 1997; Effect Design, Part 1: Reverberator and Other Filters (PDF); CCRMA, Stanford University, Stanford, CA; Paper ; Available from: https://aes.org/publications/elibrary-page/?id=10160*
- *Stewart, R., & Sandler, M. (2007). STATISTICAL MEASURES OF EARLY REFLECTIONS OF ROOM IMPULSE RESPONSES.*